

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

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COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks: 30

Annexure A

Constraint-Based Problem Solving

1. The multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

Understanding of Problem and Constraints

The main constraints are the absence of high-speed wired internet, limited budget, requirement for cloud applications and video conferencing, secure communication with headquarters, uninterrupted operation, and future scalability. Therefore, the network must provide adequate bandwidth, reliable connectivity, strong security, and backup connectivity without excessive cost.

Problem Decomposition

The problem can be divided into:

1. Internet connectivity selection
2. Network device selection
3. Security implementation
4. Reliability and backup
5. Future scalability

Proposed Solution

A **4G/5G fixed wireless connection** can be used as the primary internet connection because wired broadband is unavailable in the rural location. If 5G coverage is unavailable, a 4G LTE connection can be used.

A **business-grade router** with dual-WAN capability should be installed. It can connect the primary 4G/5G network and a secondary connection such as another mobile network. A managed switch can connect computers, printers, IP phones, and other wired devices. Wireless access points can provide Wi-Fi for employees.

For security, the router should include a **stateful firewall, VPN, NAT, and access-control rules**. A site-to-site VPN should securely connect the branch office to headquarters. WPA3 should be used for wireless authentication.

Reliability and Performance

Because business operations must continue even if the primary connection fails, dual-WAN failover should automatically switch to the backup connection. Traffic prioritization or QoS can give higher priority to video conferencing and critical cloud applications.

Cost and Scalability

Instead of deploying expensive infrastructure, the company can start with 4G/5G connectivity and scalable networking equipment. Additional access points, switches, and bandwidth can be added as the branch grows.

Conclusion

A 4G/5G fixed wireless connection with dual-WAN failover, business router, managed switch, secure Wi-Fi, firewall, and VPN provides a practical balance between bandwidth, reliability, security, cost, and scalability.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Understanding of Problem and Constraints

The university has more than 3,000 students distributed across classrooms, laboratories, libraries, and hostels. The major constraints are high simultaneous-user density, limited access points, interference, secure authentication, sufficient coverage, performance, and budget.

Problem Decomposition

The problem can be divided into:

1. User-density analysis
2. Access-point placement
3. Frequency-band selection
4. Channel planning
5. Authentication and security
6. Bandwidth management

Proposed Solution

The university should deploy Wi-Fi 6 access points because Wi-Fi 6 is designed to handle high-density environments efficiently.

Access points should not simply be placed according to building area. They should be positioned according to user density. Classrooms, laboratories, libraries, and hostels with many simultaneous users should receive higher priority. Low-density areas can use fewer access points.

A proper site survey should be performed to identify dead zones and interference. Access points should use appropriate channel planning rather than allowing neighboring access points to use overlapping channels.

Frequency Band Selection

The 5 GHz band should be preferred for high-performance applications because it provides more channels and generally experiences less congestion than 2.4 GHz. The 2.4 GHz band can be retained for devices that require greater range or compatibility.

Where supported, Wi-Fi 6 features such as OFDMA and MU-MIMO can improve efficiency when many students are connected simultaneously.

Authentication and Security

The university should use WPA3-Enterprise with 802.1X authentication connected to a central authentication server. Students and staff can authenticate using institutional credentials instead of sharing one password.

Separate SSIDs or VLANs can be created for students, faculty, guests, and administrative systems. This prevents unauthorized access to sensitive university resources.

Cost Optimization

Since the number of access points is limited by budget, APs should be concentrated in high-density locations instead of being distributed equally. Centralized wireless management can monitor AP utilization and identify overloaded areas.

Conclusion

A Wi-Fi 6, density-based campus network using 5 GHz, intelligent channel allocation, WPA3-Enterprise, 802.1X, and VLAN-based separation provides an effective balance between coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Understanding of Problem and Constraints

The financial institution handles sensitive customer information and online banking services. Therefore, the major constraints are strong cybersecurity, regulatory compliance, high performance, controlled operational costs, secure employee access, and protection against unauthorized access.

Problem Decomposition

The network can be divided into:

1. Network segmentation
2. Firewall deployment
3. Authentication
4. Intrusion detection
5. Secure access to customer data
6. Monitoring and compliance

Network Segmentation

The organization should implement VLAN-based segmentation and separate critical systems into different security zones.

For example:

- Employee VLAN
- Server VLAN
- Database VLAN
- Guest VLAN
- Management VLAN
- DMZ for externally accessible services

The database containing sensitive customer information should not be directly accessible from the employee network.

Firewall Deployment

A next-generation firewall (NGFW) should be deployed at the network perimeter. It should control incoming and outgoing traffic using security policies.

A second internal firewall can be used between sensitive server/database zones and the general employee network. This creates multiple security layers and reduces the impact of a compromised device.

Authentication

Employees should use multi-factor authentication (MFA) for sensitive systems. Role-based access control should ensure that employees can access only the information required for their jobs.

For example, a normal employee should not have administrative access to the customer database.

Intrusion Detection and Prevention

An **IDS/IPS** should continuously monitor network traffic for suspicious behavior, malware, unauthorized access attempts, and abnormal traffic patterns. Logs should be centrally collected and monitored for security incidents.

Performance and Cost

Security controls should be placed strategically rather than inspecting unnecessary traffic repeatedly. VLANs, centralized authentication, firewall policies, and automated monitoring reduce administrative effort while maintaining security.

The organization can also use existing network infrastructure where possible and prioritize security controls around the most sensitive systems.

Conclusion

A segmented network with VLANs, DMZ, NGFW, MFA, role-based access control, IDS/IPS, and centralized monitoring provides strong protection for customer data while maintaining performance, compliance, and reasonable operational costs.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand